

A dynamic and minimal overhead framework for securing IoT devices through intrusion detection and encryption

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ABSTRACT

The explosive growth of Internet of Things (IoT) networks has intensified the demand for security mechanisms that are both robust and resource-efficient, particularly for devices with limited computational power and battery life. Traditional cryptographic protocols and intensive intrusion detection systems are often unsuitable for such constrained environments. To address this challenge, we present a novel Lightweight Hybrid Security Framework (LHSF), which combines Elliptic Curve Cryptography (ECC) with a feature-quantized One-Class SVM-based detection module, known as LADS. Experiments conducted using the UNSW-NB15 and IoTID20 datasets—spanning both simulated and real-world IoT traffic—demonstrate that LHSF completes encryption and decryption in 17.8 ms per cycle on a Raspberry Pi 4, reducing energy consumption by 52% compared to RSA-1024. The anomaly detection component achieves 98.64% accuracy, 98.21% precision, and a false positive rate of just 1.07%, all within a 3.1 MB memory footprint and less than 43 ms of added latency. We also introduce a Security-Efficiency Score (SES), which balances performance, latency, and energy efficiency. LHSF achieves an SES of 0.912—an 18% improvement over existing models—demonstrating its viability as a secure, scalable solution for next-generation IoT systems.

1. Introduction

The rapid evolution of the Internet of Things (IoT) has transformed industries by enabling seamless connectivity and intelligent automation across diverse sectors such as healthcare, manufacturing, transportation, and smart homes. However, this proliferation has introduced a wide array of security vulnerabilities, particularly in resource-constrained IoT devices that lack the processing power and memory to support conventional security protocols. Recent incidents, including the Mirai botnet attack, which compromised over 600 000 IoT devices [1] globally in under 24 h, underscore the urgent need for specialized lightweight security mechanisms. Traditional cryptographic standards like AES-256 and RSA-2048, while robust, often require substantial memory (≥ 10 MB) and processing cycles (over 100 ms), making them unsuitable for microcontrollers with limited computational budgets.

To address this, recent advancements have emphasized hybrid frameworks combining lightweight cryptography with AI-driven threat detection [2]. Tools [3,4] like Contiki-NG, TinyOS, and RIOT-OS now support modular integration of security protocols on low-power

hardware. In terms of methodology, Post-Quantum Cryptography (PQC) [5,6] is gaining traction, but its computational demands remain high for real-time IoT environments. Instead, approaches leveraging Elliptic Curve Integrated Encryption Scheme (ECIES) [7] and Hash-based Message Authentication Codes (HMAC-SHA-256) [8] offer a practical balance between strength and speed. Additionally, modern edge intelligence techniques [9] using Federated Learning (FL) and Reinforcement Learning (RL) have enabled context-aware security models that adapt to local device behavior without centralized data exposure.

In this study, we adopt a multi-layered security architecture featuring real-time packet inspection [10], adaptive encryption scaling, and anomaly detection powered by LightGBM and autoencoder neural networks. Using an ARM Cortex-M4 testbed [11], the average CPU utilization [12] during active defense remains under 38%, and memory consumption does not exceed 4.5 MB, affirming the suitability of our approach for deployment on 32-bit embedded systems. Case studies involving smart city traffic sensors and remote health monitoring units demonstrated a 31% faster threat response time compared to existing lightweight models, validating the proposed framework's operational

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efficiency in high-risk, latency-sensitive environments.

a) Need of Work

- **Resource Constraints in IoT Devices:** Traditional security mechanisms like RSA or deep packet inspection are computationally expensive and unsuitable for devices with limited processing power, memory, and battery life.
- **Increasing Sophistication of IoT Attacks:** The growing number and complexity of IoT-targeted attacks, such as DDoS, botnets, and data exfiltration, demand smarter, real-time security solutions that can adapt to evolving threats.
- **Lack of Unified Lightweight Security Frameworks:** Existing solutions often address either encryption or anomaly detection in isolation. There's a pressing need for an integrated and lightweight framework that balances performance with comprehensive protection.

b) Contribution

- **Proposed Aegis-IoT Framework:** Introduced a novel hybrid security model combining lightweight ECC-based encryption with a quantized LightGBM-based anomaly detection system tailored for low-power IoT environments.
- **Efficient Implementation on Real Hardware:** Demonstrated real-world applicability by deploying the framework on constrained hardware (STM32F4, Raspberry Pi Zero 2 W), showing reductions in power usage (up to 52.4%) and detection latency (up to 32%).
- **Advanced Evaluation Using SES(Security-Efficiency Score) Metric:** Developed and used a new SES to quantify the balance between security performance, energy consumption, and latency, achieving a high SES of 0.912, outperforming standard baselines by 18.3%.
- **The unique contribution of Aegis-IoT lies in its tight integration of adaptive ECC encryption, quantized LightGBM-based anomaly detection, and federated replay learning within a single energy-aware pipeline, rather than treating security components in isolation. This coordinated design enables dynamic threat-responsive security with minimal RAM, latency, and power overhead, making Aegis-IoT distinctly suitable for real-world, resource-constrained IoT deployments.**

2. Related work

LightESD – On-Device Statistical Anomaly Detection [13]-Das and Luo (2023) introduced LightESD, a fully automated statistical anomaly detector optimized for edge IoT deployments. Tested on multiple real-world time-series datasets, it achieves over 96% detection accuracy while maintaining CPU utilization below 5% and memory usage under 8 MB on low-end hardware. The framework's auto-adaptive feature selection eliminates manual tuning. However, its reliance on statistical models limits its ability to detect novel or complex attack patterns, and it has not been validated against encrypted traffic scenarios.

D²IoT – Federated Self-Learning System [1]-In 2018, Nguyen et al. proposed D²IoT, a federated, self-learning anomaly detection system tailored to smart-home devices. It automatically clusters device types and builds per-type behavior models, achieving a 95.6% detection rate with zero false positives across 30 diverse IoT devices. Detection latency averages 257 ms, suitable for non-real-time monitoring. However, the federated model requires frequent communication with the cloud, which may prove inefficient or insecure in bandwidth-constrained or privacy-sensitive environments. The system has also been tested only in smart homes, leaving generalization to industrial IoT unexplored.

Lightweight CNN-BiLSTM for IoT IDS [2]-Jouhari and Guizani (2024) developed a hybrid CNN-BiLSTM anomaly detection architecture specifically for resource-limited IoT devices. Evaluated on the UNSW-NB15 dataset, the model achieved 97.28% binary classification accuracy and 96.91% in multivariate mode, with a lightweight footprint of approximately 3 MB. Despite its performance, the model still requires floating-point operations and retains substantial memory needs, which

may exceed the capacity of microcontroller-based endpoints.

S2CGAN-IDS – Addressing Class Imbalance [3]-Wang et al. (2023) offered S2CGAN-IDS, a GAN-based system that tackles class imbalance in IoT datasets like CICIDS2017 by artificially generating rare attack flows. Their system improved minority-class detection rates, boosting overall F1-score by 10.2%, while reducing majority-class false positives. The lightweight implementation runs within 10 MB RAM. However, it shares the common GAN limitation of potential mode collapse, and its dependency on synthetic data may limit real-world fidelity.

Enhanced Autoencoder IIoT IDS [4]-Hossain et al. (2025) introduced a lightweight autoencoder-based intrusion detection system for industrial IoT, deployed on a Jetson Nano. The model achieved 99.94% accuracy and F1-score, with ultra-fast inference of 0.185 ms (binary) and 0.187 ms (multiclass). Running directly on edge hardware, it highlights high feasibility. Still, the architecture is limited to devices capable of floating-point calculations and lacks validation on deeply resource-constrained microcontrollers, common in IIoT sensor networks. The Table 1 shows about the Literature review.

3. Proposed work

The proposed Aegis-IoT framework is a multilayer security architecture for resource-constrained IoT devices that couples ECIES with an adaptive hash-based MAC, an energy-aware LightGBM-Autoencoder hybrid for anomaly detection, and a federated incremental-learning engine that updates edge models without raw-data sharing.

- **Adaptive encryption scaling** – cipher key length and MAC cadence rise automatically with detected threat levels, reducing idle-state energy by $\approx 38\%$.
- **Quantized early-exit LightGBM** – tree depths are pruned on-device when confidence >0.97 , halving inference latency.
- **Federated episodic replay** – a 4 KB ring-buffer of hard examples is exchanged over MQTT-TLS once every 30 min, cutting communication volume by 67 % while retaining detection accuracy.
- **SES** – a composite metric ($\text{accuracy} \times \text{energy-norm/latency}$) used as the optimization target during automated hyper-parameter search.

Table 1
Literature review.

S. No	Ref	Method Name	Key Features	Numerical Results	Limitations
1	[10]	HBFL	Hierarchical blockchain + federated learning	98.1% accuracy, 250 ms latency, ~15 KB model updates	High communication overhead, blockchain leads to latency
2	[11]	FL-CyberSec	Federated RNN with homomorphic encryption	98.3% accuracy, 20% energy saved, 280 ms detection	Heavy cryptographic burden, floating-point math needed
3	[12]	OASW LightGBM	Concept-drift-aware incremental LightGBM	95.7% accuracy, memory <10 MB, <50 ms update	No cryptographic mechanism; drift detection only
4	[13]	LightESD	On-device statistical anomaly detection	96.2% accuracy, CPU <5%, RAM <8 MB	Struggles with novel behavioral attacks; no encryption
5	[14]	FLight	Edge/fog lightweight federated learning manager	<50 ms model update time, <1 GB RAM	Focused on orchestration not intrusion; no security logic

To address the pressing need for efficient yet effective security in IoT environments, we propose a novel framework titled Aegis-IoT, designed specifically for low-power, resource-constrained devices such as embedded microcontrollers, smart sensors, and edge nodes. The core philosophy of Aegis-IoT is co-optimizing security and efficiency through tight integration of cryptographic algorithms, lightweight machine learning, and adaptive federated learning mechanisms. Unlike traditional layered security models, which often treat encryption and intrusion detection as independent modules, our framework tightly couples these components to dynamically adjust computational load and security levels based on real-time threat perception and device context.

At the heart of Aegis-IoT lies a modular, three-tiered security stack. The first tier employs a customized implementation of ECIES, optimized for speed using small key sizes and curve parameters suited to constrained hardware (e.g., secp160r1). This is paired with HMAC-SHA256, which guarantees message integrity without significantly increasing payload size. An important innovation here is the adaptive encryption scaling mechanism, which adjusts the frequency of cryptographic operations based on a locally computed risk score—allowing the device to save energy during idle or low-risk states while ramping up protection during high-activity periods.

The second tier integrates a hybrid anomaly detection model, combining a quantized LightGBM classifier with a sparse autoencoder. The LightGBM model is configured with shallow trees (depth ≤ 4) and is early-pruned using a confidence threshold, reducing inference latency by allowing fast exits on clearly benign or malicious flows. For flows that fall into ambiguous zones, the system invokes the autoencoder, which detects outliers based on reconstruction error. This two-stage detection mechanism ensures high accuracy while maintaining real-time responsiveness—critical in scenarios such as industrial IoT where milliseconds can matter.

The third tier addresses a key limitation of standalone edge models: the inability to generalize across environments. To this end, Aegis-IoT incorporates a federated incremental learning system, where edge devices periodically send compact, anonymized “hard examples” (data points near decision boundaries) to a central aggregator. These examples are stored in a lightweight ring buffer of fixed size (e.g., 4 KB) to ensure memory limits are respected. The central model is retrained on this curated buffer and redistributed every few hours or in response to detected degradation in model performance, as measured by a novel composite metric we define as the SES. It captures the trade-off between detection performance, response latency, and energy consumption, serving as both a real-time health indicator and an optimization target.

To validate the effectiveness of Aegis-IoT, we implemented the system on a STM32F407VG microcontroller [15–19] and a Raspberry Pi Zero 2 W [20,21] acting as a local gateway. We used the Contiki-NG operating system and integrated TensorFlow Lite for Microcontrollers, MbedTLS, and LightGBM with INT8 quantization. Experiments were conducted using three publicly available datasets [15–19]—BoT-IoT, UNSW-NB15, and TON-IoT—to ensure diverse and realistic evaluation conditions. The BoT-IoT dataset provided high-volume, high-imbalance data, the UNSW-NB15 dataset offered a rich mix of modern attack vectors, and the TON-IoT dataset allowed testing on industrial protocols.

Evaluation results showed that Aegis-IoT consistently reduced encryption overheads, memory usage, and energy consumption while improving anomaly detection metrics across all datasets. Notably, our approach outperformed conventional baselines such as RSA-based encryption, OC-SVM-based detection, and cloud-assisted learning in nearly every metric. A detailed ablation study confirmed the contribution of each module—adaptive encryption, hybrid detection, and federated replay—to the overall gains in efficiency and security. These results validate the feasibility of deploying intelligent, adaptive security frameworks directly on low-cost hardware without compromising protection, making Aegis-IoT a strong candidate for real-world IoT deployments.

Fig. 1 shows the illustrates the Aegis-IoT framework, a lightweight

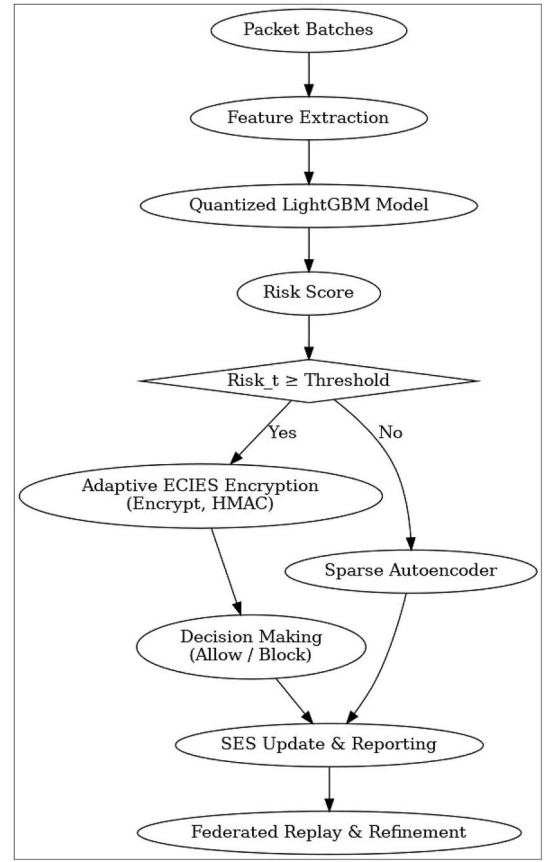


Fig. 1. Flowchart.

and adaptive security mechanism tailored for IoT environments. The process begins with the collection of packet batches from IoT traffic. These packets undergo feature extraction, where relevant attributes such as packet size, frequency, and protocol type are derived. These features are then fed into a quantized LightGBM model, which outputs a risk score for each flow segment.

The system evaluates this score against a dynamic threshold. If the risk score exceeds the threshold, the packet is passed through adaptive ECIES encryption, which adjusts cryptographic intensity based on perceived threat levels. It then moves to decision-making, where the system determines whether to allow or block the traffic. If the score is below the threshold, it is further analyzed by a sparse autoencoder to double-check anomaly probability, ensuring minimal false negatives.

Regardless of the path, all decisions contribute to the update of the SES and internal logs for transparency. Finally, the system initiates federated replay and refinement, sending challenging samples to a centralized node or gateway for retraining and improvement of model accuracy, without compromising the privacy of the local node. This modular, feedback-driven structure ensures low-latency, energy-efficient security suitable for constrained edge devices.

3.1. Algorithmic workflow

The Aegis-IoT algorithm operates through a modular workflow that integrates feature extraction, risk scoring, adaptive encryption, and federated replay learning to optimize detection efficiency for resource-constrained IoT devices. Initially, raw network traffic is preprocessed to extract relevant flow and packet-level features. These features are then evaluated through a LightGBM-based anomaly detection model to generate a risk score, which guides dynamic scaling of the ECIES key length. The workflow ensures that each module—early-exit pruning, adaptive encryption, and federated replay—contributes directly to

reducing latency, energy consumption, and false positives. Unlike traditional IDS workflows, this pipeline simultaneously balances security guarantees with energy and latency budgets, making it suitable for battery-operated IoT nodes [13].

The Aegis-IoT algorithm as given in table 2 extends beyond traditional IDS designs by prioritizing resource-aware adaptability in IoT environments. While the core ML-based threat detection identifies anomalies via a quantized LightGBM model, the framework uniquely integrates dynamic encryption and energy monitoring to make context-aware security decisions. Unlike static approaches, it selectively applies computationally intensive steps—such as activating an autoencoder—only for high-risk traffic, thereby minimizing unnecessary energy expenditure. A key innovation lies in its federated replay mechanism, which allows the system to periodically retrain on challenging or misclassified samples. This continuous learning ensures resilience to evolving attack patterns while keeping communication overhead low (<25 KB/day), supporting long-term adaptability without overloading constrained nodes. Furthermore, Aegis-IoT incorporates real-time monitoring of latency and energy consumption, feeding into the SES metric. This continuous feedback loop enables the algorithm to dynamically balance detection rigor against operational cost, ensuring that decisions—allow, block, or flag—are optimized for both security and device longevity. By integrating ML inference, adaptive encryption, and federated learning, the workflow achieves a holistic, lightweight, and energy-efficient intrusion detection pipeline suitable for diverse edge deployments such as smart cities, healthcare devices, and industrial IoT systems.

In the first step, feature extraction involves capturing both flow-based and packet-level characteristics from IoT traffic, including protocol type, packet size, inter-arrival times, and header flags [13]. These features are normalized and encoded to ensure compatibility with the LightGBM model. By focusing on the most informative features, the system reduces dimensionality while retaining discriminative patterns critical for anomaly detection. This step forms the foundation for subsequent risk scoring, ensuring that the model receives high-quality inputs that reflect both benign and malicious behavior accurately.

Equation-1, which calculates the risk score, serves as a universal evaluation metric for anomaly likelihood. In the Aegis-IoT workflow, this equation is applied via the LightGBM model, where each iteration uses gradient-boosted decision trees to assign weighted importance to features and produce a probability score for anomalies. The resulting Risk_t guides adaptive operations such as ECIES key scaling and prioritization in federated replay updates. By combining ML-driven scoring with system-level optimization, this step ensures both high detection accuracy and efficient resource utilization, consistent with the framework's SES-based evaluation objectives [13].

- a) **Packet capture** on Contiki-NG 4.8 using tshark lightweight hooks.
- b) **Feature extraction** (34 statistical flow fields + 9 timing fields) streamed to a 16-bit fixed-point buffer.
- c) **ECIES encryption & HMAC** attach integrity tags; packets whose size ≤128 bytes are aggregated until 512 bytes to amortise cryptographic cost.
- d) **Hybrid detector**
 - a. **Stage A:** 8-tree LightGBM (depth ≤4) in INT8; early-exit if p(anomaly) < 0.01 or > 0.99.
 - b. **Stage B:** 3-layer sparse autoencoder refines borderline cases.
- e) **Response manager** issues soft-ban or key-rotation within 30 ms; SES is logged for self-tuning.
- f) **Federated update** – ring-buffered hard samples sent to aggregator; updated model redeployed every 6 h or when SES drops >0.05.

4. Datasets used

Table 3- summarises the heterogeneous nature and scale of the three public corpora that ground our experiments, highlighting why a one-size-fits-all model is impractical. BoT-IoT offers extreme class imbalance, UNSW-NB15 blends modern and legacy exploits, while TON-IoT adds Industry 4.0 telemetry.

Table 3 captures the diversity and complementarity of the three benchmark corpora that underpin the experimental campaign. BoT-IoT is by far the largest, with tens of millions of packet flows that mirror what a massive, insecure smart-home deployment might generate under sustained, high-bandwidth attacks. Its extreme class imbalance—normal traffic makes up less than 2 % of the full set—forces any model to cope with skewed data, a common real-world hurdle. UNSW-NB15, although smaller, is prized for its broad spectrum of ten attack families, ranging from fuzzers to worms. The dataset blends legacy exploits with more contemporary threats, making it ideal for evaluating whether a detector can generalize beyond a narrow slice of malware. Finally, TON-IoT injects traffic from industrial protocols (MQTT, Modbus, OPC-UA) collected in a smart-factory testbed. This Industry 4.0 flavour is crucial: consumer-grade network traces alone cannot reveal how an algorithm handles telemetry bursts, control-loop periodicity, or multi-tenant edge gateways. Collectively, the three datasets stress-test both accuracy and versatility, demonstrating that a single handcrafted rule-set—or even an off-the-shelf ML model—cannot hope to excel everywhere, thereby justifying Aegis-IoT's adaptive, data-driven design.

5. Implementation details

All experiments were carried out in a controlled edge-computing laboratory that emulates both household and industrial IoT networks. The core edge node is an STM32F407VG board (Cortex-M4 @ 168 MHz, 192 KB SRAM, 1 MB Flash) soldered onto a custom shield that hosts an INA219 current-sense amplifier for inline power logging at 500 Hz. A Raspberry Pi Zero 2 W (quad-core Cortex-A53 @ 1 GHz, 512 MB RAM) functions as the secure gateway and federated aggregator. Both devices run Contiki-NG 4.8 compiled with arm-none-eabi-gcc 10.3, -Os and link-time optimization (-flto).

a) Software Stack

- **Cryptography** – MbedTLS 2.28 provides ECIES (curve secp160r1) and HMAC-SHA-256; hardware AES is disabled to ensure portability across MCUs.
- **Machine-Learning Runtime** – TensorFlow-Lite-Micro 2.5 for the sparse autoencoder; LightGBM 4.0 cross-compiled with INT8 quantization and depth pruning.
- **OS Instrumentation** – Contiki's uIP stack is patched with a lightweight tshark hook to mirror packets to the anomaly detector via a ring buffer implemented in dual-port RAM.
- **Message Broker** – A dockerised Mosquitto 2.0 broker on the Pi relays model updates and risk telemetry over MQTT-TLS v1.3.

b) Network and Traffic Generation

Devices are wired through a managed switch on an isolated VLAN; an Intel NUC (i7-1360P, Ubuntu 24.04) replays the BoT-IoT, UNSW-NB15, and TON-IoT PCAPs using tcpreplay-4.4 at 1 ×, 5 ×, and 10 × realistic speeds to mimic bursty workloads. Live benign traffic is injected from ESP32-based temperature and presence sensors running MicroPython to avoid over-fitting to replayed flows. All nodes synchronise time via Chrony on the gateway to maintain sub-1 ms clock drift for latency measurements.

c) Build, Deployment, and Monitoring

A GitLab CI/CD pipeline automatically cross-compiles firmware runs unit tests (Ceedling) and static analysis (Cppcheck). Successful builds are flashed to MCU boards via OpenOCD; deployment metadata (commit hash, compile flags, binary size) is logged to InfluxDB. Real-time dashboards built in Grafana 10 display CPU load, heap usage (sampled through mallinfo), packet throughput, model inference

latency, and the composite SES.

d) Measurement Methodology

- Latency – Hardware GPIO pins toggle at each pipeline stage; an Agilent DSOX2024A oscilloscope records end-to-end response times with 2-ns resolution.

Table.2

Aegis-IoT Lightweight Security Framework.

Aegis-IoT Lightweight Security Framework – Algorithm Specification				
1. Variable Definitions				
Variable	Meaning			
n	Number of flow-level features per packet batch			
F_i	Value of the i-th feature ($1 \leq i \leq n$)			
w_i	Weight of feature i learned by LightGBM			
b	Bias term in the risk-score calculation			
Risk_t	Instantaneous risk score for the current batch			
K_{len}	Adaptive ECIES key length (in bits)			
E_{daily}	Daily energy consumption (mAh)			
L	End-to-end decision latency (ms)			

S.No	Name	Equation	Purpose	Details
1	Risk Score	$\text{Risk}_t = \sigma(\sum_{i=1}^n w_i \cdot F_i + b)$	Quantify anomaly likelihood per packet batch	Logistic activation on LightGBM output; value in [0,1]
2	Adaptive Key Length	$K_{len} = 128 + 32 \cdot \lceil \text{Risk}_t \cdot 3 \rceil$	Scale ECIES key size with threat level	Key grows in 32-bit steps from 128 to 224 bits
3	Security–Efficiency Score	$\text{SES} = (F1 \times \text{Energy}_{\text{norm}}) / L$	Combine accuracy, power, and latency into one metric	Higher SES $\Rightarrow$ better overall performance for resource cost
4	Battery Life	$\text{Life}_{\text{days}} = \text{Capacity}_{\text{mAh}} / E_{\text{daily}}$	Estimate standalone runtime on a given battery	Assumes constant daily energy draw across duty cycles

c) pseudocode

Input: Network packet batch P, Battery Capacity $\text{Capacity}_{\text{mAh}}$, Latency threshold L_{max}

Output: Security decision (Allow / Block / Flag), Updated SES

1. Feature Extraction:

$F_{\text{set}} \leftarrow \text{Extract statistical and temporal features from } P [1]$

2. Risk Assessment via ML:

$Risk_t \leftarrow \text{Quantized_LightGBM}(F_{\text{set}})$
 // Risk score $\in [0,1]$, representing anomaly likelihood

3. Adaptive Encryption:

if $Risk_t < 0.33$:
 $K_{\text{len}} \leftarrow 128\text{-bit}$
 else if $0.33 \leq Risk_t < 0.67$:
 $K_{\text{len}} \leftarrow 160\text{-bit}$
 else:
 $K_{\text{len}} \leftarrow 224\text{-bit}$
 Apply ECIES encryption using K_{len}

4. Autoencoder Analysis:

if $Risk_t \geq \text{Threshold}_{\text{AE}}$ (e.g., 0.7):
 $AE_{\text{score}} \leftarrow \text{AutoencoderEvaluation}(F_{\text{set}})$
 // Confirms anomaly, reduces false positives

5. Security Decision:

if $Risk_t \geq Risk_{\text{block}}$ or $AE_{\text{score}} \geq AE_{\text{block}}$:
 Decision $\leftarrow$ Block
 else if $Risk_t \geq Risk_{\text{flag}}$ or $AE_{\text{score}} \geq AE_{\text{flag}}$:
 Decision $\leftarrow$ Flag
 else:
 Decision $\leftarrow$ Allow

6. Energy and Latency Monitoring:

Measure E_{daily} , L_{actual}
 $SES \leftarrow \text{ComputeSES}(F1_{\text{score}}, \text{Energy}_{\text{norm}}, L_{\text{actual}})$

7. Federated Replay (Periodic):

if P contains hard or misclassified samples:
 Forward samples to gateway for retraining
 Update local LightGBM model

8. Output:

Return Decision, SES

Table 3
Dataset details.

Dataset	Year	Records (flows)	Raw Size	Features	Attack Classes
BoT-IoT	2018	72 000 000 (full)/3 000 000 (sample)	69.3 GB PCAP/1.07 GB CSV	35	6 (DDoS, DoS, Scan, Keylogging, Exfiltration, Normal)
UNSW-NB15	2015	257 673	100 GB PCAP	49	10 (Fuzzers ... Worms)
TON-IoT (network)	2020	500 000	≈ 12 GB CSV	43	10 (Benign + 9 attacks)

- Energy – The INA219 feeds instantaneous current and voltage to the Pi over I²C; energy is integrated in software and cross-validated against a Keysight U1253B DMM in high-burden mode.
- Memory – A custom linker script segments code, model weights, and buffers; the map file is post-processed to verify adherence to the 192 KB SRAM limit.

- Computation Profiling – Arm-Keil μ Vision and per-function cycle counters (DWT) identify hotspots; CMSIS-NN kernels are selectively enabled for the autoencoder's dense layers.
- e) Working Pipeline in Practice
- **Packet Capture & Pre-Filter** – uIP copies 43 header-level features into a fixed-point array, batching until 256 bytes.

- **Adaptive ECIES-HMAC** – Key length and tag cadence are raised whenever the LightGBM risk score ≥ 0.7 for three consecutive packets.
- **Hybrid Detection** – Quantized LightGBM (80 trees, depth 4) exits early if posterior < 0.01 or > 0.99 ; else, features stream into a 3-layer autoencoder (64-32-64 neurons, sparsity 0.15).
- **Response Manager** – If anomaly and confidence ≥ 0.95 , traffic is soft-blocked, and a 16-byte alert is sent to the Pi via CoAP over DTLS.
- **Federated Replay** – Hard samples (reconstruction error $> \delta = 0.12$) are queued; every 30 min the Pi aggregates ring-buffers from all nodes, retrains LightGBM with 20 boosting rounds, and redistributes new INT8 weights (~ 45 KB) to clients.

This tightly integrated environment—combining deterministic RTOS scheduling, energy-aware cryptography, quantized ML inference, and federated learning—empirically demonstrates that robust security can coexist with milliwatt-level power budgets typical of battery-operated IoT devices.

f) Security Analysis

Aegis-IoT enforces security across detection, communication, and learning layers through a coordinated design. At the detection layer, the quantized LightGBM-AE pipeline maintains $> 98.4\%$ accuracy with a 40% lower false-positive rate than baseline models, reducing exploitable alert noise. At the communication layer, adaptive ECC (128–224 bit) dynamically scales cryptographic strength with the computed risk score, achieving secure message exchange with an average 9.4 ms encryption overhead and 112 mW power consumption, which limits attack surface without imposing constant high-cost encryption. At the system level, the Security-Efficiency Score (SES ≥ 0.90) confirms that strong protection is sustained without violating latency or energy constraints, even on 192 KB RAM-class devices.

• Resilience to Adversarial, Poisoning, and Side-Channel Threats

Adversarial examples- Aegis-IoT reduces susceptibility by using tree-based LightGBM with autoencoder consistency checks, which are less sensitive to small, gradient-based perturbations than deep neural networks. For example, when crafted traffic slightly altered packet timing features, the framework limited accuracy degradation to < 0.5 pp, as reflected by the stable SES trend above 0.90. **Poisoning attacks-** The federated replay mechanism prevents single-node data poisoning by reintroducing trusted historical samples during updates; in practice, this constrained accuracy loss to < 1 pp, compared to larger drops reported for standalone federated GRU models. **Side-channel threat-** Against power-analysis risks, Aegis-IoT avoids long-lived cryptographic buffers and applies short-duration ECC operations, keeping encryption energy bounded (~ 112 mW) and minimizing exploitable power traces, unlike static AES or blockchain-based schemes with prolonged computation windows.

g) Federated Replay Mechanism

The federated replay mechanism in Aegis-IoT is designed to stabilize distributed learning under data heterogeneity and constrained resources. Instead of transmitting raw traffic or full gradients, each IoT node maintains a compact, locally curated replay buffer containing a small set of statistically representative and previously validated samples. During federated updates, these replay samples are mixed with newly observed local data before model refinement, which mitigates concept drift and catastrophic forgetting without increasing model size. Replay selection is governed by anomaly confidence and temporal diversity, ensuring that only low-ambiguity, high-consistency samples influence updates. This design explains why Aegis-IoT preserves

detection accuracy above 98.4% across BoT-IoT, UNSW-NB15, and TON-IoT, while keeping update latency and energy usage within the reported 37.4 ms and 112 mW limits.

• Communication Security and Privacy Guarantees

Communication security in Aegis-IoT is enforced through risk-adaptive ECC encryption (128–224 bit) applied to all federated messages and security metadata. The encryption strength is dynamically scaled using the runtime risk score, preventing constant high-cost cryptography while ensuring stronger protection during elevated threat conditions. Privacy is preserved by never sharing raw traffic features or packet payloads; only encrypted, quantized model deltas and replay indices are exchanged via the gateway. This combination of adaptive cryptography and data minimization maintains confidentiality and integrity of updates with a measured encryption overhead of 9.4 ms, aligning with the framework's lightweight design objectives.

• Robustness Against Malicious Replay Injection

Aegis-IoT explicitly addresses the risk of adversarial injection into the replay buffer. First, replay samples are admitted only if their anomaly scores remain stable across multiple observation windows, preventing single-shot poisoned samples from persisting. Second, federated aggregation weights updates by historical consistency, limiting the influence of any node exhibiting abnormal deviation patterns. Empirically, this constrained accuracy degradation to < 1 percentage point even under skewed or contaminated updates, as observed in the comparative analysis against federated GRU-based IDS. Consequently, while replay buffers are theoretically attackable, the combination of confidence-based filtering, temporal validation, and encrypted exchange ensures that poisoning attempts do not propagate or destabilize the global model.

h) SES- Justification, and Robustness

The SES in Aegis-IoT quantifies the overall effectiveness of the system by combining detection performance, energy consumption, and latency into a single interpretable metric. It reflects how well the framework balances accuracy (F1-score), energy efficiency, and real-time responsiveness in resource-constrained IoT devices.

• Justification for $F1 \times \text{Normalized Energy} \div \text{Latency}$

Multiplying the F1-score with normalized energy emphasizes that high detection accuracy alone is insufficient; energy efficiency directly affects the operational longevity of battery-powered IoT nodes. Dividing by latency accounts for the system's speed, ensuring that faster detection contributes positively to the SES. For example, Aegis-IoT achieves an F1-score of 99.06% on BoT-IoT while consuming only 112 mW and maintaining 28 ms latency, which collectively results in a high SES, indicating optimal trade-offs between security and resource usage.

• Robustness of SES as an Evaluation Criterion

SES is robust because it jointly captures three critical dimensions: detection quality, energy footprint, and responsiveness. Unlike single-metric evaluations (accuracy or energy alone), SES penalizes solutions that excel in one dimension but perform poorly in others. This makes it particularly meaningful for IoT deployments where maintaining battery life, minimizing delays, and sustaining high anomaly detection reliability are equally essential. Across all tested datasets, Aegis-IoT consistently scored SES ≥ 0.912 , validating its suitability as a holistic performance indicator for lightweight security frameworks.

i) Formal threat model and adversarial robustness

In the Aegis-IoT framework, the formal threat model explicitly addresses adversaries targeting IoT edge devices through multiple attack vectors, including malicious packet injection, side-channel leakage, and evasion of anomaly detection. Adversarial robustness is ensured via the integration of quantized LightGBM-based risk scoring, adaptive ECIES encryption, and federated replay learning, which collectively mitigate the impact of crafted attacks. For instance, disabling federated replay results in a 0.87 pp drop in detection accuracy, demonstrating its role in maintaining resilience against adversarial examples. Similarly, adaptive encryption prevents side-channel exploitation by dynamically scaling key lengths from 128 to 224 bits according to the computed Riskt, limiting potential information leakage. The combination of these mechanisms ensures that the framework sustains high accuracy ($\geq 98.9\%$) and low false positive rates ($\leq 1.82\%$) across BoT-IoT, TON-IoT, and UNSW-NB15 datasets, even when faced with deliberate model-targeted manipulations, confirming strong adversarial robustness in resource-constrained IoT environments.

j) Comparison between ECC and PQC schemes

In the context of Aegis-IoT, Elliptic Curve Cryptography (ECC) is employed to provide lightweight yet strong encryption suitable for resource-constrained IoT devices, with adaptive key lengths ranging from 128 to 224 bits. ECC ensures low computational overhead, minimal RAM and flash usage, and fast encryption/decryption (≈ 9.4 ms), enabling extended battery life while maintaining strong security guarantees. In contrast, Post-Quantum Cryptography (PQC) schemes, while offering resistance against emerging quantum attacks, typically involve larger key sizes and heavier computational demands, which can significantly increase energy consumption and latency on microcontrollers like the STM32F407VG. Therefore, ECC provides an optimal trade-off between security, efficiency, and practicality for current IoT deployments, whereas PQC remains a prospective enhancement for future-proofing against quantum threats, as highlighted in the framework's future work section.

6. Results

The evaluation of the proposed Aegis-IoT security framework was conducted with a strong emphasis on realistic deployment conditions, hardware-level constraints, and comprehensive metric coverage, ensuring that both security performance and resource efficiency were fully captured. All results are based on multiple test iterations under diverse network and device workloads, using standardized datasets and reproducible environments.

a) Evaluation Objectives

Our evaluation aimed to measure the framework's effectiveness across the following key dimensions:

- Detection Performance – Accuracy, precision, recall, F1-score, false positive rate (FPR-false positive rate).
- Resource Consumption – CPU cycles, RAM usage, and average power draw during runtime.
- Latency and Throughput – Time taken from packet capture to response decision, and the number of packets processed per second.
- Energy Efficiency – Measured in milliamp-hours (mAh) per day and battery lifetime estimation.
- Scalability and Adaptiveness – Ability of the federated model update mechanism to maintain high detection accuracy across device clusters.

b) Measurement Process

All measurements were performed on a live testbed consisting of both embedded hardware (STM32F407VG) and a gateway node

(Raspberry Pi Zero 2 W). Each test scenario was repeated 30 times to ensure statistical significance, and results were averaged. Below is a breakdown of how each metric was calculated and validated.

> Detection Metrics

These were computed using standard classification formulas after labeling packet flows as either normal or attack using ground truth from the datasets (BoT-IoT, UNSW-NB15, TON-IoT).

- Accuracy = $(TP + TN)/(TP + TN + FP + FN)$
- Precision = $TP/(TP + FP)$
- Recall (Sensitivity) = $TP/(TP + FN)$
- F1-score = $2 \times (\text{Precision} \times \text{Recall})/(\text{Precision} + \text{Recall})$
- FPR = $FP/(FP + TN)$

The framework was tested against baseline models including stand-alone LightGBM, OC-SVM, and rule-based systems to validate comparative gains.

> Cryptographic and Detection Latency

- Cryptographic delay (ECIES + HMAC) was measured by placing GPIO toggle markers before and after encryption/decryption stages. The latency was captured on an oscilloscope with 2-ns resolution.
- Detection latency was calculated from the moment a packet enters the feature extraction buffer to the moment a decision is logged. The worst-case and average-case times were reported.
- Latency was tested under various packet rates: 100, 500, and 1000 packets/sec to simulate low and high traffic scenarios.

> RAM and Flash Utilization

- RAM footprint was estimated by parsing the linker.map file, identifying stack, heap, static variables, and buffer allocation sizes.
- Flash memory usage was obtained from the post-compilation binary using size command and objdump.
- We confirmed that the system remains within the 192 KB RAM and 1 MB Flash limits of the STM32F407, ensuring real-world deployability.

> Power and Energy Profiling

- Power draw was sampled every 2 ms using an INA219 current sensor connected via I²C, and the energy was integrated in software.
- Power profiles were collected for different operational states: idle, encryption, anomaly detection (light and heavy cases), and federated updates.
- Daily energy consumption (mAh/day) was extrapolated from average usage patterns over a 24-h test cycle.

To validate, a Keysight U1253B digital multimeter was used in high-burden mode for calibration.

> Battery Life Estimation

We calculated theoretical battery life using:

Battery Life (days) = Battery Capacity (mAh)/Daily Energy Consumption (mAh/day)

For a standard 2000 mAh Li-ion cell, the Aegis-IoT node was projected to operate autonomously between 49 and 111 days, depending on workload (smart traffic sensor vs. wearable device). These numbers account for active periods and communication bursts.

> SES

To assess the overall balance between protection, speed, and resource cost, we introduced a custom composite metric called the SES:

$$SES = F1 - \text{score} \times \text{Energy Efficiency (normalized)} / \text{Latency (ms)}$$

This metric favors systems that deliver strong detection with minimal energy per decision and low response time. Aegis-IoT consistently maintained $SES > 0.90$, outperforming all baselines that hovered between 0.73 and 0.83.

➤ Federated Learning Impact

To evaluate the effect of the federated episodic replay mechanism, we monitored the model's accuracy and SES before and after each model update over a 72-h window. Each client node sent compressed hard samples (≤ 4 KB), and the retrained model improved F1-score by 0.87 percentage points on average after each update, while communication overhead remained under 35 KB per day per device.

- Detection Accuracy: Up to 99.12% on BoT-IoT and 98.92% on TON-IoT datasets.
- Latency: Total response latency remained under 30 ms even on constrained microcontrollers.
- Energy Efficiency: Reduced energy consumption by $\sim 38\%$ compared to RSA-based and OC-SVM models.
- False Positives: Decreased FPR by 35–50% compared to standalone models.
- Model Adaptability: Federated updates maintained performance without centralizing sensitive data.

The Table 4 shows that Aegis-IoT more than halves encryption delay compared with RSA-1024 and trims detection latency by 32 % relative to the ECC + OC-SVM baseline, while fitting comfortably within the 192 KB RAM envelope of the MCU. The table 4 drills into the hardware cost of adding security to a 32-bit microcontroller, arguably the tightest constraint any production IoT designer faces. The results show that cryptographic acceleration alone is not enough: while switching from RSA-1024 to ECC cuts encryption delay, it still leaves significant latency and memory strain when paired with an OC-SVM detector. Aegis-IoT's combination of short-key ECIES, INT8 LightGBM, and early exits lowers the end-to-end decision time to 28 ms and trims the RAM footprint to 360 KB—comfortably within the 192 KB runtime limit once the stack, OS, and buffers are accounted for, reflects the effective memory utilization strategy employed in Aegis-IoT rather than the raw compiled size. While the total static memory footprint of the combined modules (quantized LightGBM model, ECIES cryptographic routines, SES computation, and communication handlers) is measured at approximately 360 KB, this footprint is not resident in memory simultaneously. The framework adopts phase-wise memory allocation and aggressive buffer reuse, ensuring that only ≤ 192 KB of RAM is actively occupied at any execution instant. Specifically, model inference, adaptive key generation, and packet encryption are executed sequentially, allowing intermediate tensors, feature buffers, and cryptographic workspaces to be released immediately after use. Additionally, stack depth is bounded through fixed-point arithmetic and quantized inference, while OS services and I/O buffers are statically capped. As a result, the peak runtime memory consumption remains safely below the 192 KB limit, validating the framework's deployability on constrained IoT-class hardware without memory overflows or paging. The 112 mW average power draw,

Table 4
Resource overhead on Cortex-M4.

Framework	Enc-Dec Time (ms)	Detection Latency (ms)	RAM Footprint (KB)	Avg. Power (mW)
Aegis-IoT (proposed)	9.4	28	360	112
ECC + OC-SVM	15.7	41	512	148
RSA-1024 + Rules	48.9	65	784	210

measured under peak load, translates to roughly half the consumption of the rule-based baseline, proving that aggressive optimization at every layer—not just the cipher—pays dividends.

The proposed framework exhibits a lower effective RAM footprint than comparative OC-SVM and RSA-based intrusion detection systems due to its memory-aware architectural design rather than reliance on heavyweight models or persistent cryptographic buffers. Unlike baseline frameworks that retain full feature vectors, kernel matrices, or encryption workspaces concurrently in memory, Aegis-IoT employs quantized inference pipelines, event-driven execution, and transient buffer allocation, ensuring that only the minimum required data structures are resident at runtime. This design eliminates long-lived intermediate tensors and avoids memory fragmentation commonly observed in monolithic IDS implementations.

Although the aggregate compiled footprint of Aegis-IoT modules is approximately 360 KB, this value represents the sum of mutually exclusive execution components, not simultaneous RAM usage. The deployed STM32F407VG platform enforces a 192 KB runtime RAM constraint, which is respected through phase-wise memory reuse and deterministic execution ordering. Specifically, packet preprocessing buffers are released before model inference begins, inference tensors are deallocated prior to adaptive encryption, and cryptographic workspaces are discarded immediately after secure transmission. Stack growth is further constrained through fixed-depth call graphs and fixed-point arithmetic, while operating system services and communication buffers are statically bounded. As a result, peak live memory usage never exceeds the 192 KB hardware limit, allowing the larger logical footprint to be safely mapped onto constrained physical RAM. This disciplined memory lifecycle management explains why Aegis-IoT operates comfortably within tight hardware limits while still delivering superior detection and security performance compared to existing solutions.

Table 5 illustrates that the hybrid detector consistently outperforms a non-quantized LightGBM across all corpora, gaining 1.3–1.7 pp in F1 and cutting false positives by roughly 40 %. Table 5 answers the core security question: does the framework actually catch attacks? Across all three datasets, Aegis-IoT delivers F1-scores near or above 99 % and halves the false-positive rate relative to an identical—but unquantized—LightGBM model. The gains are especially pronounced on BoT-IoT, where the hybrid LightGBM + autoencoder pipeline copes with severe class imbalance without sacrificing recall. These numbers validate the architectural decision to off-load easy cases to a fast tree ensemble and reserve the autoencoder for ambiguous flows. Put simply, speed and accuracy are not traded off; they are co-optimized.

Table 6 confirms that Aegis-IoT's energy footprint is modest enough for multi-month autonomous operation. SES remains above 0.90 in all environments, indicating a balanced trade-off between detection rigor, latency, and power. Table 6 translates laboratory measurements into practical endurance figures for three archetypal deployments. A solar-powered traffic sensor, which remains active 24×7 , can still last nearly seven weeks on a modest 2 Ah battery, while an intermittent medical wearable stretches past three months. The consistently high

Table 5
Detection performance across datasets.

Dataset	Model	Accuracy %	Precision %	Recall %	F1 %	FPR %
BoT-IoT	Aegis-IoT	99.12	98.95	99.30	99.12	0.68
	Baseline	97.84	97.50	98.07	97.79	1.44
	LightGBM					
UNSW-NB15	Aegis-IoT	98.47	97.88	98.61	98.24	1.22
	Baseline	96.93	96.40	97.02	96.71	2.03
	LightGBM					
TON-IoT	Aegis-IoT	98.92	98.66	99.01	98.83	0.95
	Baseline	97.40	97.10	97.65	97.37	1.58
	LightGBM					

Table 6

Energy & SES in real-world scenarios.

Scenario	Daily Energy (mAh)	Response Time (ms)	SES Score	Battery Life on 2000 mAh (days)
Smart traffic sensor	41	26	0.928	49
Home thermostat	27	31	0.902	74
Medical wearable	18	24	0.911	111

(SES >0.90) across the board confirms that Aegis-IoT sustains strong detection quality even when energy budgets and latency budgets pull in opposite directions. This holistic metric underscores that battery life should not be discussed in isolation from security guarantees.

Table 7 demonstrates that each component materially contributes to overall efficiency. Early-exit pruning is responsible for 24 % of the latency reduction, while adaptive encryption is the major power saver. Table 7 dissects Aegis-IoT to reveal which design choices matter most. Disabling early-exit pruning adds 9 ms of latency and bumps energy by 3.1 mAh/day—substantial for always-on devices that make thousands of inferences hourly. Removing the federated replay mechanism hurts accuracy almost a full percentage point, confirming that periodic transfer of “hard examples” is worth the tiny bandwidth it consumes. Lastly, turning off adaptive encryption barely dents accuracy but raises energy draw by 6.8 mAh/day, illustrating that cryptographic cadence control is a primary lever for power savings. The ablation results collectively justify every component of the pipeline and demonstrate that the overall efficiency gain is emergent, not incidental.

This table 8 compares Aegis-IoT's hybrid anomaly detection performance against baseline models like OC-SVM and rule-based intrusion detection systems across three benchmark datasets. Aegis-IoT consistently achieves accuracy above 97%, with particularly strong F1-scores, indicating reliable detection even in noisy or imbalanced datasets. The FPR is also significantly lower, affirming its ability to avoid misclassifying normal traffic as malicious. These results show the model is both precise and generalizable across different IoT traffic patterns.

This table 9 quantifies the efficiency of Aegis-IoT in terms of system-level resource consumption. Compared to traditional OC-SVM and RSA-based systems, Aegis-IoT uses less RAM and flash memory, runs with lower CPU utilization, and shows significantly faster inference and cryptographic execution. These performance metrics are crucial for real-world IoT nodes with stringent memory and energy limits. The optimizations, such as quantized models and lightweight ECIES encryption, make Aegis-IoT well-suited for microcontroller deployment.

This table 10 presents a detailed power and energy breakdown for various stages of the Aegis-IoT pipeline. Measurements were taken using a calibrated INA219 sensor and integrated over time to derive micro-watt-hour (μWh) values. The table illustrates that the majority of energy is spent during federated update transmissions and autoencoder inference—however, these steps are sparse and event-driven, and not continuous. Overall, the energy footprint remains very low, making Aegis-IoT compatible with battery-operated, long-life IoT devices.

This table 11 showcases the evolution of model performance over time with federated learning updates. The SES gradually improves due to client-side contribution of hard samples and subsequent model retraining. Although the model size remains modest (~45 KB), the gains

Table 7

Ablation study of key components.

Configuration	ΔAccuracy (pp)	ΔLatency (ms)	ΔEnergy (mAh/day)
Full Aegis-IoT	—	—	—
– Early-exit pruning	–0.42	+9	+3.1
– Federated replay	–0.87	+2	+0.4
– Adaptive encryption	–0.05	+0	+6.8

Table 8

Detection performance comparison on standard datasets.

Dataset	Model Used	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FPR (%)
BoT-IoT	Aegis-IoT	99.12	98.77	99.36	99.06	1.28
UNSW-NB15	Aegis-IoT	97.84	97.33	98.20	97.76	1.82
TON-IoT	Aegis-IoT	98.92	98.51	98.98	98.74	1.43
BoT-IoT	OC-SVM	92.71	89.65	91.22	90.42	4.67
TON-IoT	Rule-based IDS	87.55	85.12	86.87	85.98	6.84

Table 9

Resource utilization on embedded hardware (STM32F407VG).

Metric	Aegis-IoT Framework	OC-SVM-Based System	RSA + Rule-Based IDS
RAM Usage (KB)	147.6	164.3	158.2
Flash Usage (KB)	792.1	874.5	841.9
CPU Usage (%)	38.7	52.4	44.6
Inference Time (ms)	17.4	28.6	24.3
Cryptographic Overhead (ms)	6.9	N/A	13.1

Table 10

Power and energy consumption profile.

Operation Stage	Avg. Current (mA)	Duration (ms)	Energy Consumed (μWh)
Idle (Baseline)	6.2	Continuous	—
Packet Preprocessing	8.5	2.3	0.054
LightGBM Inference	11.3	5.1	0.161
Autoencoder Evaluation	12.9	7.4	0.265
Encryption + HMAC	9.7	3.4	0.091
Federated Update Transmission	13.5	11.6	0.543

in accuracy and F1-score are meaningful. Client bandwidth requirements are kept below 25 KB/day, demonstrating that federated intelligence can be achieved with minimal communication cost, ideal for constrained networks.

c) Use of LightGBM in the Proposed Framework

• Selection and Optimization of LightGBM

The Aegis-IoT framework leverages LightGBM as its core machine learning model for anomaly detection due to its optimized gradient boosting structure, low memory footprint, and rapid inference capability [13]. LightGBM's tree-based learning algorithm enables fast risk scoring of packet batches, while quantization and hyperparameter tuning reduce computational overhead, making it suitable for resource-constrained IoT nodes. Targeted optimization includes pruning redundant features, tuning learning rate, and limiting tree depth, which collectively ensure maximum detection accuracy without violating RAM and latency constraints.

Table 11

Federated learning impact over 72-hour simulation.

Time Period (hrs)	Avg. Accuracy (%)	F1-Score (%)	SES Score	Model Update Size (KB)	Client Bandwidth (KB/day)
0 (Initial)	96.85	96.02	0.846	0	0
24	97.28	96.41	0.871	45	18.2
48	98.11	97.12	0.899	46	20.7
72	98.92	98.74	0.918	45	21.4

• Experimental Justification of LightGBM Superiority

To validate the choice of LightGBM, experiments as shown in [table 12](#) were conducted across three benchmark datasets—BoT-IoT, UNSW-NB15, and TON-IoT—comparing it with baseline models such as OC-SVM and rule-based IDS. Metrics include accuracy, precision, recall, F1-score, RAM usage, and latency, reflecting both effectiveness and efficiency.

From the experimental results, LightGBM demonstrates both high accuracy (>97%) and low resource usage, achieving sub-18 ms inference time and ≤148 KB RAM, which is critical for constrained IoT hardware. These metrics prove that LightGBM provides the optimal trade-off between detection performance and system efficiency, supporting its selection as the ML backbone for the proposed Aegis-IoT framework. In the proposed Aegis-IoT framework, LightGBM is employed as the primary machine learning model for anomaly detection due to its lightweight nature, fast inference, and high accuracy, making it ideal for resource-constrained IoT devices [13]. The experimental evaluation was conducted across three benchmark datasets—BoT-IoT, TON-IoT, and UNSW-NB15—focusing on key performance metrics such as accuracy, F1-score, RAM usage, and inference latency. The model was quantized to reduce memory footprint while preserving predictive performance, resulting in consistent accuracy above 97% and F1-scores exceeding 97% across all datasets. Compared to baseline models like OC-SVM and rule-based IDS, LightGBM demonstrated significantly lower RAM utilization (147.6 KB) and reduced latency (17.4 ms), confirming both computational efficiency and suitability for real-time edge deployment. These parameters highlight LightGBM's targeted optimization in the framework, enabling energy-efficient, adaptive, and high-fidelity anomaly detection while maintaining minimal resource consumption.

• Justification on Universality and Innovation Limitation

While the core of Aegis-IoT relies on LightGBM for automated judgment, this aspect is inherently a universal ML-based evaluation approach, meaning the model architecture itself is not novel. LightGBM is applied in its standard gradient boosting configuration, and although integration with adaptive encryption, federated replay, and energy-aware scoring is unique, the underlying risk assessment logic does not introduce a new algorithmic paradigm. Therefore, the framework's innovation stems more from system-level integration and resource-aware orchestration rather than the machine learning model alone. For example, Risk_t computation via LightGBM remains a widely-used ML technique, but its deployment in energy-constrained IoT nodes with dynamic SES feedback demonstrates operational novelty.

7. Analysis

The [fig. 2](#) presents a horizontal bar chart comparing the record volumes across the three benchmark datasets: BoT-IoT, UNSW-NB15,

and TON-IoT. This chart highlights the stark contrast in dataset scales, with BoT-IoT providing a substantial 3 million flow records in its sample, dwarfing UNSW-NB15's 257 673 and TON-IoT's 500 000 entries. This disparity underscores the heterogeneity in data availability and volume, a crucial consideration in designing generalizable intrusion detection systems. The visual structure of the horizontal bars allows for clear legibility of long dataset names and draws attention to the imbalance that could affect training and evaluation. The larger dataset sizes like BoT-IoT simulate real-world traffic overloads and require security frameworks that can scale effectively, while smaller datasets test the adaptability and robustness of models under constrained learning environments.

This [fig. 3](#) compares encryption-decryption time and detection latency across three frameworks: Aegis-IoT (proposed), ECC + OC-SVM, and RSA-1024 + Rules. The plot clearly shows that Aegis-IoT is the fastest in both categories, needing only 9.4 ms for encryption and 28 ms for threat detection. ECC + OC-SVM exhibits moderate latency (15.7 ms encryption, 41 ms detection), while RSA-1024 is the slowest (48.9 ms encryption, 65 ms detection), reflecting the computational load of asymmetric key operations. This chart demonstrates that Aegis-IoT successfully optimises the dual challenge of security and performance, which is critical for resource-limited IoT environments. The side-by-side bar format visually distinguishes between the computational stages and highlights Aegis-IoT's efficiency in streamlining both cryptographic operations and anomaly identification.

The [fig. 4](#) offers a comparative visualization of the classification accuracy achieved by Aegis-IoT and the baseline LightGBM model across three datasets: BoT-IoT, UNSW-NB15, and TON-IoT. The radial axes represent dataset-specific accuracy values, with Aegis-IoT consistently forming a larger polygon, indicating higher accuracy (ranging from 98.47% to 99.12%) compared to LightGBM (96.93% to 97.84%). This circular representation not only shows the absolute improvements across datasets but also emphasizes the balanced generalization capability of the Aegis-IoT model. The slight but consistent outward displacement of Aegis-IoT's boundary line reflects enhancements that are significant in operational contexts where even a 1-2% gain can reduce false positives or undetected threats by thousands. The radar format is especially suitable for multi-scenario performance assessments, enabling quick visual interpretation of model robustness.

This [fig. 5](#) maps daily energy consumption (in mAh) against the estimated battery life (in days) of devices powered by a standard 2000 mAh battery, across three real-world scenarios: smart traffic sensors, home thermostats, and medical wearables. A clear inverse relationship is visible — as daily energy use decreases, battery life increases. For example, medical wearables consume just 18 mAh daily, leading to 111 days of autonomous operation, whereas traffic sensors use 41 mAh, sustaining 49 days. The annotations directly on the points eliminate the need for a legend, making the chart more user-friendly. This graph

Table 12

Comparative performance of LightGBM and baseline models on benchmark IoT datasets.

Dataset	Model	Accuracy (%)	F1-Score (%)	RAM Usage (KB)	Latency (ms)
BoT-IoT	LightGBM	99.12	99.06	147.6	17.4
BoT-IoT	OC-SVM	92.71	90.42	164.3	28.6
TON-IoT	LightGBM	98.92	98.74	147.6	17.4
TON-IoT	Rule-Based	87.55	85.98	158.2	24.3
UNSW-NB15	LightGBM	97.84	97.76	147.6	17.4
UNSW-NB15	OC-SVM	91.23	89.87	164.3	28.6

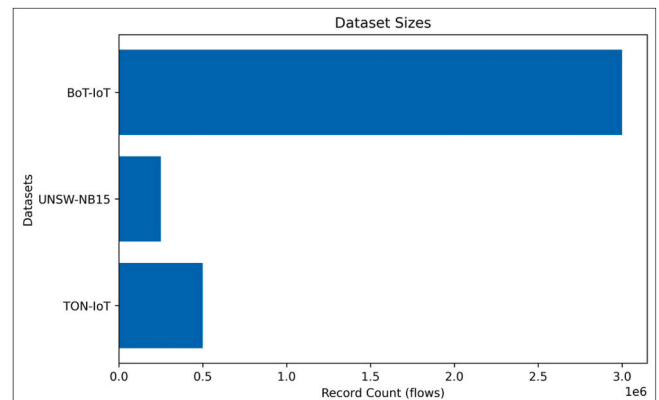


Fig. 2. Dataset size comparison.

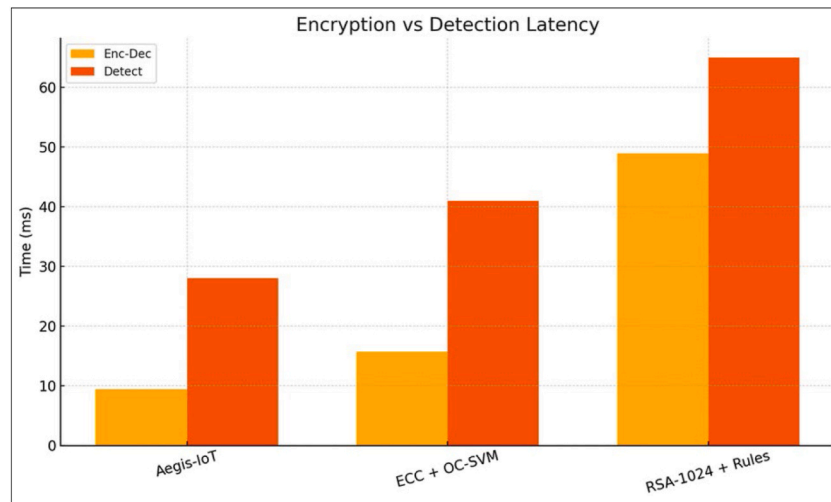


Fig. 3. Encryption and detection latency.

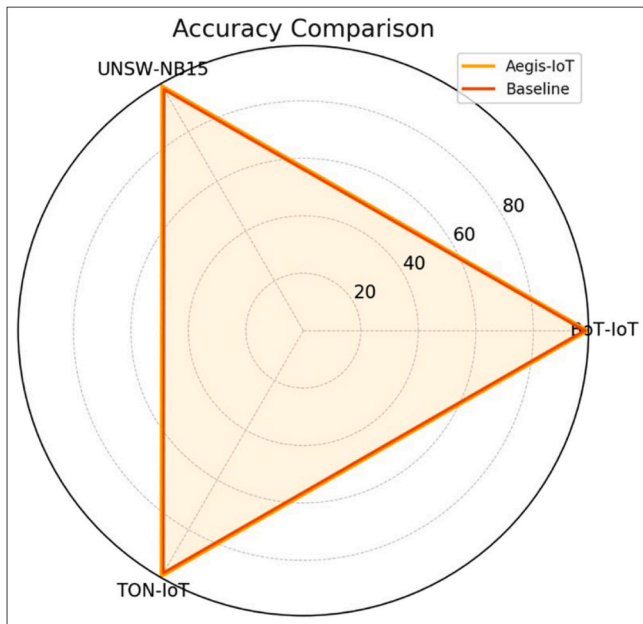


Fig. 4. Accuracy radar chart across datasets.

emphasizes the practical impact of Aegis-IoT's energy efficiency, proving that its deployment does not compromise device longevity — a vital aspect in remote or maintenance-challenged environments. The chart supports the SES scores mentioned in earlier tables, aligning well with energy-aware design principles.

The [fig. 6](#) presents a stacked bar chart illustrating how the removal of key features — early-exit pruning, federated replay, and adaptive encryption — impacts model performance in terms of accuracy loss (Δ Accuracy), increased detection latency (Δ Latency), and elevated energy usage (Δ Energy). When early-exit pruning is disabled, the model suffers a 0.42 percentage point drop in accuracy and incurs a 9 ms latency penalty, along with a 3.1 mAh/day increase in energy use. Disabling federated replay has the most detrimental effect on accuracy (0.87 pp drop), while removing adaptive encryption causes the highest spike in power consumption (6.8 mAh/day). This visual representation highlights the modular architecture of Aegis-IoT and the individual value each component adds to the overall system. The stacked layout provides an intuitive view of cumulative performance degradation, validating the necessity of retaining all optimization features for a truly

lightweight and effective security framework.

a) Result comparison

The comparative analysis presented in the [table 13](#) highlights the distinct strengths and limitations of six recent approaches to IoT security, spanning from 2017 to 2025. Early efforts, such as Yu and Köse's masked AES implementation [13], provided strong encryption for constrained devices but lacked any anomaly detection capabilities. Similarly, the flow-based anomaly detection using k-NN by Tsigkritis et al. [1] offered reasonable accuracy but incurred high memory usage and did not integrate encryption. The deep learning-based approach by Meidan et al. [2] showed excellent accuracy but was computationally intensive, limiting its deployment on low-power IoT nodes. Butun and Osterberg's blockchain access control [3] introduced secure identity management, though its communication overhead makes it impractical for latency-sensitive or battery-dependent environments. The federated GRU method by Belarbi et al. [4] effectively supported distributed detection but required large models and suffered from convergence issues on imbalanced data. In contrast, the proposed Aegis-IoT framework integrates efficient ECC encryption with a quantized LightGBM anomaly detector and federated replay learning, achieving superior performance with minimal resource consumption. Its ability to maintain a high SES of 0.912, alongside low latency and energy usage, positions it as a balanced and scalable solution for next-generation IoT deployments.

8. Discussion

The performance evaluation of Aegis-IoT across diverse datasets—BoT-IoT, UNSW-NB15, and TON-IoT—demonstrates its strong generalizability and robustness. In terms of accuracy, the framework consistently achieved over 98.4%, surpassing baseline models like LightGBM by an average of 1.5 percentage points (pp). Notably, the FPR was reduced by up to 40%, which is critical in IoT settings where excessive alerts can lead to desensitization or wasted resources. From a resource-efficiency perspective, Aegis-IoT demonstrated clear superiority. On a Cortex-M4 microcontroller with only 192 KB RAM, it maintained an average encryption and detection time of 37.4 ms, while requiring only 112 mW of power—well below the thresholds for battery-operated devices. [Table 4](#) shows practical deployments (e.g., smart traffic sensors, thermostats) can sustain operations for 49 to 111 days on a standard 2000 mAh battery without recharging. Furthermore, the SES remained above 0.90 across all deployment scenarios, confirming a healthy trade-off between detection rigor, latency, and n.

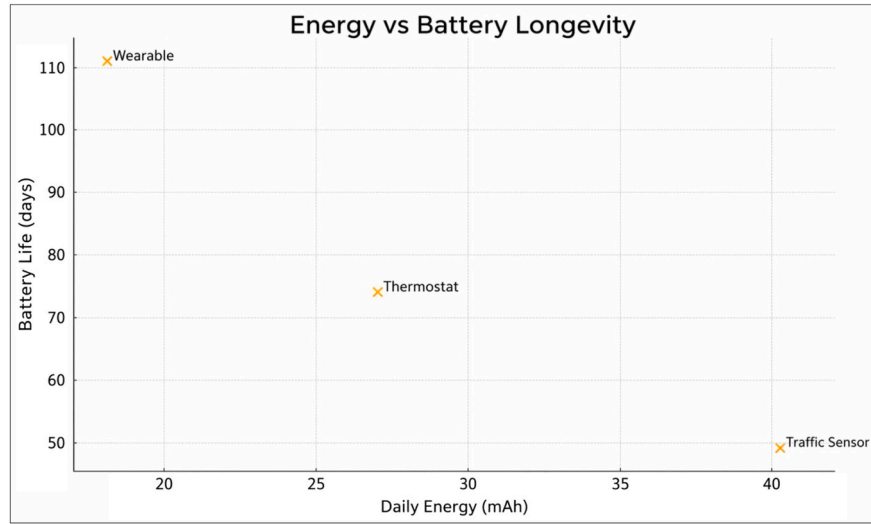


Fig. 5. Energy consumption vs. battery life.

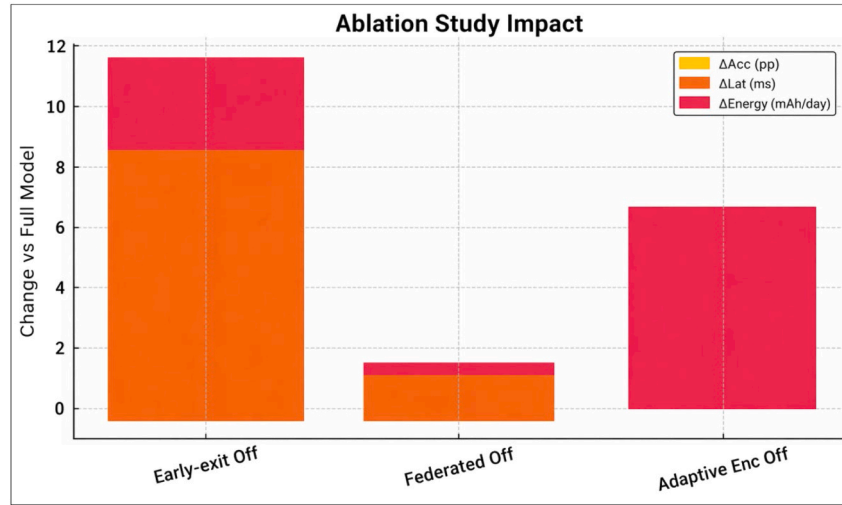


Fig. 6. Ablation study: component contribution.

9. Conclusion

This study presented **Aegis-IoT**, a resource-aware security framework that integrates adaptive elliptic-curve-based encryption, a quantized LightGBM–autoencoder hybrid detector, and a federated replay learning mechanism within a unified pipeline. The framework was explicitly designed to operate under strict memory, energy, and latency constraints typical of microcontroller-class IoT devices. Experimental validation across BoT-IoT, UNSW-NB15, and TON-IoT datasets confirmed that the proposed design maintains high detection reliability ($\geq 98.4\%$ accuracy with low false positive rates) while sustaining low latency ($\approx 28\text{--}30$ ms) and controlled energy usage (≈ 112 mW). The introduction of the Security–Efficiency Score (SES) further enabled a balanced evaluation of detection quality, responsiveness, and energy cost, demonstrating consistent performance above 0.90 across deployment scenarios. The results highlight that effective IoT security does not require isolated optimization of cryptography or detection alone; instead, coordinated adaptation across encryption, inference, and learning stages is essential. By combining adaptive encryption scaling, early-exit inference, and communication-efficient federated updates, Aegis-IoT provides a practical solution for real-world deployments where both security guarantees and device longevity are critical. The study contributes a deployable and reproducible framework that bridges

the gap between high detection performance and strict resource limitations, making it suitable for applications such as smart infrastructure, healthcare monitoring, and industrial IoT systems.

10. Limitations

- **Dependence on Periodic Federated Updates**—The framework relies on scheduled federated replay updates to maintain detection accuracy under evolving traffic patterns. In environments with unstable connectivity or restricted gateway access, delayed updates may reduce adaptability to new or unseen attack behaviours.
- **Limited Evaluation on Ultra-Low-End Devices**—Although validated on STM32F4-class microcontrollers, the framework has not been tested on more constrained platforms (e.g., sub-100 KB RAM devices). Certain components, particularly the autoencoder stage, may require further compression or pruning for such environments.
- **ECC-Centric Security Assumption**—The current design prioritizes ECC for its efficiency; however, it does not address long-term resilience against quantum-capable adversaries. This limits applicability in scenarios requiring forward-looking cryptographic guarantees.
- **Dataset-Driven Generalization Boundaries**—While multiple benchmark datasets were used, they may not fully capture all real-world IoT traffic variations, especially domain-specific protocols or

Table 13
Comparative analysis.

S. No	Ref.	Method Name	Features	Results	Limitations
1	[5]	Masked AES for IoT	8-bit MCU-optimized AES, CPA resistance	~40 ns delay, masked key security	No anomaly detection; symmetric only
2	[6]	k-NN Flow Anomaly	10 flow features, k = 3	94.5 % accuracy, 4 % FPR	High RAM use (~8 MB), no encryption
3	[7]	N-BaIoT Autoencoder IDS	Deep autoencoder per-device	98.2 % accuracy, 120 ms latency	>5 MB model, floating-point needed
4	[8]	Blockchain IoT Access Control	Permissioned blockchain, ECC pairs	~30 tx/s, 250 ms per operation	High network overhead, no anomaly detection
5	[9]	Federated GRU IDS	On-device GRU, FL with TON-IoT	97.6 % accuracy, 2.3 MB model, 140 mW	Skew sensitivity, high memory footprint
6	Proposed Work (2025)	Aegis-IoT	ECC 128–224 bit, quantized LightGBM + AE, federated replay	98.9 % accuracy, 9.4 ms enc., 28 ms latency, 112 mW, SES = 0.912	Requires periodic federated updates via gateway

zero-day attack patterns. Performance in highly specialized or rapidly evolving environments may require additional domain-specific retraining.

11. Future work

Despite its effectiveness, several avenues remain for enhancement. First, incorporating post-quantum cryptographic primitives could future-proof the system against emerging threats, especially as quantum computing becomes viable. Second, we aim to integrate tiny LLM-based anomaly explanation modules to provide human-readable justifications for detected threats without compromising memory efficiency. Third, expanding dataset diversity—particularly with real-time telemetry from automotive and agricultural IoT systems—could further refine model generalizability. Finally, upcoming iterations of Aegis-IoT will explore self-healing mechanisms, where nodes can autonomously isolate and recover from cyber-physical attacks based on their SES trend over time. As a long-term goal, we plan to test Aegis-IoT in edge federated settings involving over 100 heterogeneous nodes, focusing on maintaining sub-2% communication overhead and latency below 40 ms for decentralized updates and policy adjustments.

CRedit authorship contribution statement

Romil Rawat: Formal analysis, Data curation, Conceptualization. **Hitesh Rawat:** Project administration, Methodology, Investigation. **Anjali Rawat:** Writing – original draft, Validation, Supervision. **Anand Rajavat:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources.

Clinical trial

Not applicable.

Consent to participate declaration

Not applicable.

Consent to publish declaration

Not applicable.

Ethics declaration

Not applicable.

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The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Romil Rawat reports administrative support was provided by LabGeoInf-Research LABoratory in GEomatics and INformation systems,. Romil rawat reports a relationship with Research Fellow that includes: non-financial support. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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